Canonical averaging and the Gaussian sector of Newton preparations

Result, 2026-10-27; conditional construction. Independent copies, canonical mixing, forgetting unused copies and either weak or W_2-closure do force Gaussian preparations: every admissible finite-covariance state has an admissible Gaussian with the same mean and covariance. If, in addition, the hypotheses of the existing cusp preparation theorem hold, the one-pair Gaussian sector is exactly

$$\mathcal{G}_{\kappa_*} = \{\mathcal{N}(\mu, \Sigma) : \ \det \Sigma \geq \kappa_*^2\}, \qquad \kappa_* > 0. \tag{1}$$

The infimum κ_* of covariance actions is attained by a Gaussian. Every Gaussian above the floor can be built from two independently prepared squeezed copies of a floor state and a canonical mixer.

These conclusions are conditional on the preparation and copying premises stated below. The mathematical Gaussian covariance condition is the same shape as a quantum uncertainty condition, but its identification with $\hbar/2$ is **not** proved. The existing bounded-density cusp model fails even the two-copy mixer test. An entropy-floor class below realizes actual nonlinear cusp transport and the mixer rules with W_2-closure, but it is not weakly closed. Without cusp compatibility, copy/mixing closure allows a zero-floor Gaussian theory. Selective recordings are a separate closure question.

The central limit theorem and Gaussian symplectic covariance are standard. In quantum optics, the known Cushen–Hudson central limit has precisely an n-splitter interpretation; see Becker–Datta–Lami–Rouzé (abstract and introduction). The present calculation is a classical probability argument, not that quantum theorem. The restricted Liouville/Gaussian comparison is also established in Bartlett–Rudolph–Spekkens (abstract). The purpose here is to resolve the conditional preparation question posed in STATE, without claiming literature priority.

1. The premises and the action units

A one-pair preparation is a probability measure on $(q, p) \in \mathbb{R}^2$. Let $\mathcal{S}_1$ be nonempty and contain at least one state with finite covariance. Assume:

1. **Affine canonical covariance.** Every affine symplectic image of a member of $\mathcal{S}_1$ belongs to $\mathcal{S}_1$.
2. **A closure rule.** Either every probability-valued weak limit of members belongs to $\mathcal{S}_1$, or every W_2 limit in the finite-second-moment space $\mathcal{P}_2$ of members of $\mathcal{S}_1 \cap \mathcal{P}_2$ belongs to $\mathcal{S}_1$. The second condition is weaker: W_2 convergence implies weak convergence, but not conversely.
3. **Independent copies and canonical mixing.** For every finite collection of admissible one-pair states, independent product preparation, a joint canonical transformation of the form described below, and forgetting all but one output pair produce a member of $\mathcal{S}_1$. In particular this holds for arbitrarily many identical copies and for the two-copy 50:50 mixer.

No selective measurement or conditioning is included in item 3. The unretained output pairs may be kept as a full record before taking the marginal; the joint transformation is invertible.

Choose fixed units $L_0, P_0 > 0$ and let

$$x = q/L_0, \quad y = p/P_0, \quad S_0 = L_0 P_0. \tag{2}$$

Write $z = (x, y)$, $C = \mathrm{Cov}(z)$, and $\Omega = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. A real 2×2 map in the calibrated coordinates is canonical when its determinant is one. The physical covariance action is

$$\kappa(\mu) = \sqrt{\det \mathrm{Cov}(q, p)} = S_0 \sqrt{\det C}. \tag{3}$$

Translations and determinant-one maps do not change κ.

2. A central limit implemented by canonical transformations

Let $\mu \in \mathcal{S}_1$ have mean m and finite covariance C. Prepare n independent copies $Z_i = (x_i, y_i)$ with this law. Choose a real orthogonal matrix O_n whose first row is $(1, \dots, 1)/\sqrt{n}$ and transform

$$x' = O_n x, \qquad y' = O_n y. \tag{4}$$

This is canonical: it preserves the symplectic form because $O_n O_n^T = I$. The first output pair is

$$Z_1' = \frac{1}{\sqrt{n}} \sum_{j=1}^{n} Z_j. \tag{5}$$

Translate this pair by $(1 - \sqrt{n})m$, and forget the others. By the premises its law belongs to $\mathcal{S}_1$. Its characteristic function is

$$\left[e^{-ik\cdot m/\sqrt{n}}\, \widehat{\mu}(k/\sqrt{n}) \right]^n e^{ik\cdot m}. \tag{6}$$

Finite covariance gives the expansion $e^{-ik\cdot m/\sqrt{n}} \widehat{\mu}(k/\sqrt{n}) = 1 - \frac{1}{2n} k^T C k + o(n^{-1})$. Thus (6) tends to $\exp(ik \cdot m - \frac{1}{2} k^T C k)$. Lévy's continuity theorem and weak closure show

$$\boxed{\mathcal{N}(m, C) \in \mathcal{S}_1.} \tag{7}$$

This includes a degenerate Gaussian if C is singular. The output means and covariance are independent of n, so the second moments also converge to those of the Gaussian. Thus the convergence holds in W_2 as well: W_2 metrizes weak convergence plus convergence of second moments; see Raginsky–Sason, Lemma 3.4.3 (statement read). Either closure rule therefore admits (7). The covariance is the covariance of the original state: the theorem creates a Gaussian sector, not an action scale.

The normalization in (5) is essential. If instead one forms the arithmetic averages $\bar{x} = n^{-1} \sum x_i$ and $\bar{y} = n^{-1} \sum y_i$, then $\{\bar{x}, \bar{y}\} = 1/n$ in normalized canonical brackets. The canonically conjugate variable to $\bar{x}$ is $\sum_i y_i$, not $\bar{y}$. For independent identical copies their variances scale as $1/n$ and n, respectively, so their covariance action is not reduced merely by renaming both averages as a canonical pair.

3. A floor state and every Gaussian above it

Define

$$\kappa_* = \inf_{\mu \in \mathcal{S}_1,\ \mathrm{Cov}\,\mu < \infty} \sqrt{\det \mathrm{Cov}(q, p)}. \tag{8}$$

Suppose **in addition** that $\kappa_* > 0$. Let $\zeta_* = \kappa_*/S_0$. For every sequence μ_j with covariance actions decreasing to κ_*, (7) makes its corresponding Gaussians admissible. Their calibrated covariances C_j are positive definite. The canonical matrix

$$A_j = (\det C_j)^{1/4} C_j^{-1/2}, \qquad \det A_j = 1, \tag{9}$$

takes C_j to $\sqrt{\det C_j}\, I$. Translate the means to zero. The resulting admissible Gaussian laws converge both weakly and in W_2 to

$$G_* = \mathcal{N}(0, \zeta_* I). \tag{10}$$

Either closure rule admits G_*, which attains the floor.

Now squeeze two **independent** copies of G_* in opposite directions:

$$C_1 = \zeta_* \begin{pmatrix} e^{2r} & 0 \\ 0 & e^{-2r} \end{pmatrix}, \qquad C_2 = \zeta_* \begin{pmatrix} e^{-2r} & 0 \\ 0 & e^{2r} \end{pmatrix}. \tag{11}$$

Each squeeze is symplectic. Mix the two copies canonically by

$$(x_+, y_+) = \frac{(x_1, y_1) + (x_2, y_2)}{\sqrt{2}}, \quad (x_-, y_-) = \frac{(x_1, y_1) - (x_2, y_2)}{\sqrt{2}}, \tag{12}$$

and keep the plus pair. Its covariance is

$$C_+ = \frac{C_1 + C_2}{2} = \zeta_* \cosh(2r)\, I. \tag{13}$$

Every isotropic covariance νI, $\nu \geq \zeta_*$, is obtained by a choice of r. If $C > 0$ and $\sqrt{\det C} = \nu$, then $C = \nu B B^T$ for $B = C^{1/2}/\sqrt{\nu}$ with $\det B = 1$. A further canonical transformation and translation therefore produce every Gaussian with $\det C \geq \zeta_*^2$. Conversely no finite-covariance admissible state can have a smaller determinant by (8). Consequently

$$\{\text{Gaussian laws in } \mathcal{S}_1\} = \{\mathcal{N}(m, C) : \det C \geq \zeta_*^2\}. \tag{14}$$

The copy and mixer assumptions do real work in (11)–(13): canonical squeezing alone preserves determinant and would only produce the boundary $\det C = \zeta_*^2$. This gives the full Gaussian sector without assuming closure under arbitrary additive noise. It does not show that all non-Gaussian states above the same floor are admissible.

4. What the cusp adds, and what remains unproved

The earlier cusp theorem supplies $\kappa_* > 0$ if, in addition to affine canonical covariance and weak closure, preparation evolution is reversible and agrees with both incoming trajectories at its specified non-Lipschitz merger. Its proof also works under the weaker W_2 closure. Indeed, if $\kappa_* = 0$, the centered, canonically squeezed states constructed in equations (9)–(11) of the cusp note have second moments tending to zero, so their W_2 distance to δ_0 tends to zero. Closure admits δ_0, and the existing merger argument contradicts reversibility. Substitution in §3 proves (1), and strengthens that theorem's previous conclusion in this **enlarged premise set**: the positive infimum is attained and the whole one-pair Gaussian sector is known.

This does not prove that the premises are jointly physically realized. Indeed the earlier bounded-density classical cusp model $\mathcal{S}_M = \{0 \leq f \leq M,\ \int f = 1\}$ **fails** the two-copy premise. Let E be a centrally symmetric disk of canonical area $1/M$ and let $f = M 1_E$. For two independent copies, the retained $z_+ = (z_1 + z_2)/\sqrt{2}$ has continuous density

$$f_+(z) = 2(f * f)(\sqrt{2}z), \qquad f_+(0) = 2 \int f(w) f(-w)\, dw = 2M. \tag{14a}$$

It exceeds M on a neighbourhood of zero. Thus an explicit cusp-compatible witness for the earlier theorem is not a witness for the stronger premise set here. Its disk has the earlier sharp action $1/(4\pi M)$, unchanged by this identical-copy averaging, so the failure is the pointwise density ceiling, not a change in covariance.

The earlier cusp premise, literally stated, constrains Newtonian evolution only **if point preparations are admitted**. A positive-floor Gaussian family admits no points, so identity evolution on that family satisfies the point-compatibility clause vacuously while also satisfying the copying assumptions. The abstract combined assumptions are therefore consistent, but that example does not transport nonpoint states by Newton's cusp flow. Section 6 gives a class with actual cusp transport and the copy rules under W_2-closure; its ordinary weak closure fails. A nonlinear cusp flow does not generally preserve Gaussian shape; the theorem does not assume that it does. Under the literal earlier premises the cusp argument supplies positivity; the central limit and mixing supply the Gaussian sector. Identifying the resulting κ_* with the real-time quantum phase constant remains a further physical question.

This loss of Gaussian shape can be seen directly. For an initial centered product Gaussian with variances σ_q^2, σ_p^2, and a Newton force $F(q) = -V'(q)$, the Liouville equation gives, at almost every point,

$$\partial_t \log \rho_t(q,p)|_{t=0} = \frac{qp}{m\sigma_q^2} + \frac{pF(q)}{\sigma_p^2}. \tag{14b}$$

For the non-affine cusp force this is not a quadratic polynomial in (q,p), while the tangent to a differentiable Gaussian family has quadratic logarithmic derivative. Thus Gaussian preparations alone are not invariant under the actual cusp transport.

For comparison, the statistical-speed theorem obtains its affine-generator bound on a restricted Gaussian sector with supplied $h_* = 2\zeta$. Equation (14) gives that Gaussian covariance family under the additional cusp and copying assumptions, with $\zeta = \kappa_*$ in physical action units. It does not supply the recorded-instrument closure required by that theorem, nor does it extend the affine-generator estimate to every nonlinear generator.

5. Two closure tests

Zero branch. For every $\zeta \geq 0$ and every number n of canonical pairs, let $\mathcal{G}_\zeta^{(n)}$ be all Gaussian probability measures whose covariance satisfies

$$\Sigma + i\zeta\Omega_n \succeq 0, \qquad \Omega_n = \Omega \oplus \cdots \oplus \Omega. \tag{15}$$

These are classical probability measures, including degenerate ones when $\zeta = 0$. Here Σ and ζ refer to the calibrated coordinates of (2); the physical covariance action is $S_0\zeta$. Independent products give block-diagonal covariance; affine symplectic transformations preserve (15); and a marginal is a principal submatrix, so it also satisfies (15). Probability-valued weak limits remain Gaussian: tightness bounds the means and covariance matrices of a Gaussian sequence, and subsequential parameter limits preserve (15). Thus all the operations of §§1–2 are available for every $\zeta \geq 0$. In particular $\zeta = 0$ includes point measures and has no positive floor. This family does not establish compatibility with the cusp flow, and it does not show that copying selects any particular positive value. The zero-floor example is not a classification of every possible zero-floor class: all Dirac measures alone also satisfy the stated copying and weak-closure rules without containing nondegenerate Gaussians.

Selective records. For $\zeta > 0$, take the classical Gaussian $G_* = \mathcal{N}(0, \zeta I)$. An ideal, non-disturbing exact q-measurement yields a conditional state supported on a fixed q-line, with zero covariance determinant. It is outside $\mathcal{G}_\zeta^{(1)}$. Forgetting unused canonical output pairs is therefore much weaker than closure under arbitrary selective records. An instrument preserving a positive floor must restrict the record, introduce appropriate back-action, or impose another physical rule.

A covariance floor alone is not weakly closed. With $\zeta > 0$,

$$\mu_n = (1 - 1/n)\delta_0 + (1/n)\,\mathcal{N}(0, n\zeta I) \implies \delta_0, \qquad \mathrm{Cov}(\mu_n) = \zeta I. \tag{16}$$

A vanishing fraction of mass can carry all of the variance. Thus bounding the covariance determinant for each member does not by itself make the class weakly closed; even uniformly bounded second moments do not give the uniform integrability needed for covariance convergence. The Gaussian class in the zero-branch test has extra structure.

6. An entropy floor with actual cusp transport

There is a classical realization of the combined preparation, copy and actual cusp-transport rules if the closure rule is W_2 rather than unrestricted weak closure. In the calibrated coordinates of (2), fix any **supplied** $\zeta > 0$ and consider

$$\mathcal{E}_\zeta = \left\{ \rho(z)\,dz \in \mathcal{P}_2(\mathbb{R}^2) : h(\rho) := -\int \rho \log \rho \, dz \geq \log(2\pi e\zeta) \right\}. \tag{17}$$

The entropy threshold uses dimensionless coordinates. A finite second moment bounds differential entropy from above; the displayed lower bound makes it finite. The Gaussian $\mathcal{N}(0, \zeta I)$ belongs, so the class is nonempty. Affine canonical transformations preserve area and hence entropy and membership in $\mathcal{P}_2$.

For the bounded cusp force of the earlier theorem, the point flow is an invertible, area-preserving group outside an invariant null set. It transports every density in (17), preserving its entropy; the bounded force and finite time also preserve its second moment. Coupling the transported state to the same initial point and using dominated convergence gives continuity in W_2 with time. This is actual Liouville transport of the nonpoint states, not the vacuous identity evolution discussed in §4.

For independent $Z_i \sim \rho_i \in \mathcal{E}_\zeta$ and real coefficients a_i with $\sum_i a_i^2 = 1$, the two-dimensional entropy power inequality gives

$$N\left(\sum_i a_i Z_i\right) \geq \sum_i a_i^2 N(Z_i) \geq \zeta, \qquad N(Z) := \frac{e^{h(Z)}}{2\pi e}. \tag{18}$$

See Toscani, §6 (entropy-power inequality statement read). In particular the output of every copy-index orthogonal mixer and marginal in §2 remains in $\mathcal{E}_\zeta$. This establishes those mixers; arbitrary nonlinear joint canonical maps are not asserted.

The class is W_2-closed. Let $g(z) = (2\pi)^{-1} e^{-|z|^2/2}$. For any absolutely continuous $\mu = \rho dz$,

$$D(\mu \| g\, dz) = -h(\rho) + \log(2\pi) + \frac{1}{2}\int |z|^2\, d\mu(z). \tag{19}$$

If $\mu_n \in \mathcal{E}_\zeta$ tends to μ in W_2, their second moments converge and relative entropy is weakly lower semicontinuous. The right side of (19) is uniformly bounded, so μ is absolutely continuous and (19) gives $h(\mu) \geq \log(2\pi e\zeta)$. The Gaussian maximum-entropy inequality then gives

$$\sqrt{\det \mathrm{Cov}(Z)} \geq \frac{e^{h(Z)}}{2\pi e} \geq \zeta. \tag{20}$$

Its Gaussian sector is exactly $\det C \geq \zeta^2$, and its physical floor is $S_0\zeta$.

Ordinary weak closure is stronger and fails here. Put

$$\mu_n = (1 - 1/n)\mathcal{N}(0, \zeta e^{-2n} I) + (1/n)\mathcal{N}(0, \zeta e^{2n^2} I). \tag{21}$$

Entropy concavity gives $h(\mu_n) \geq \log(2\pi e\zeta) + 2$. Nevertheless μ_n converges weakly to δ_0, which is not in $\mathcal{E}_\zeta$; the second moments diverge, so this is not a W_2-convergent sequence. The entropy floor, like the earlier density ceiling, is an explicitly supplied classical preparation constraint. It does not identify the floor with quantum $\hbar/2$ or establish closure under arbitrary selective records.

7. Consequence for STATE

The proposed central-limit step is settled under explicit copy/mixer, discard and either weak- or W_2-closure premises. Together with the earlier point-trajectory compatibility hypotheses, it gives an attained positive covariance floor and exactly the one-pair Gaussian sector above it. The entropy model realizes actual nonlinear cusp transport and the specified mixers under W_2 closure, while demonstrating that ordinary weak closure is a separate, stronger condition. The supplied entropy floor does not derive $\hbar$ or establish selective record closure or independent action necessity.